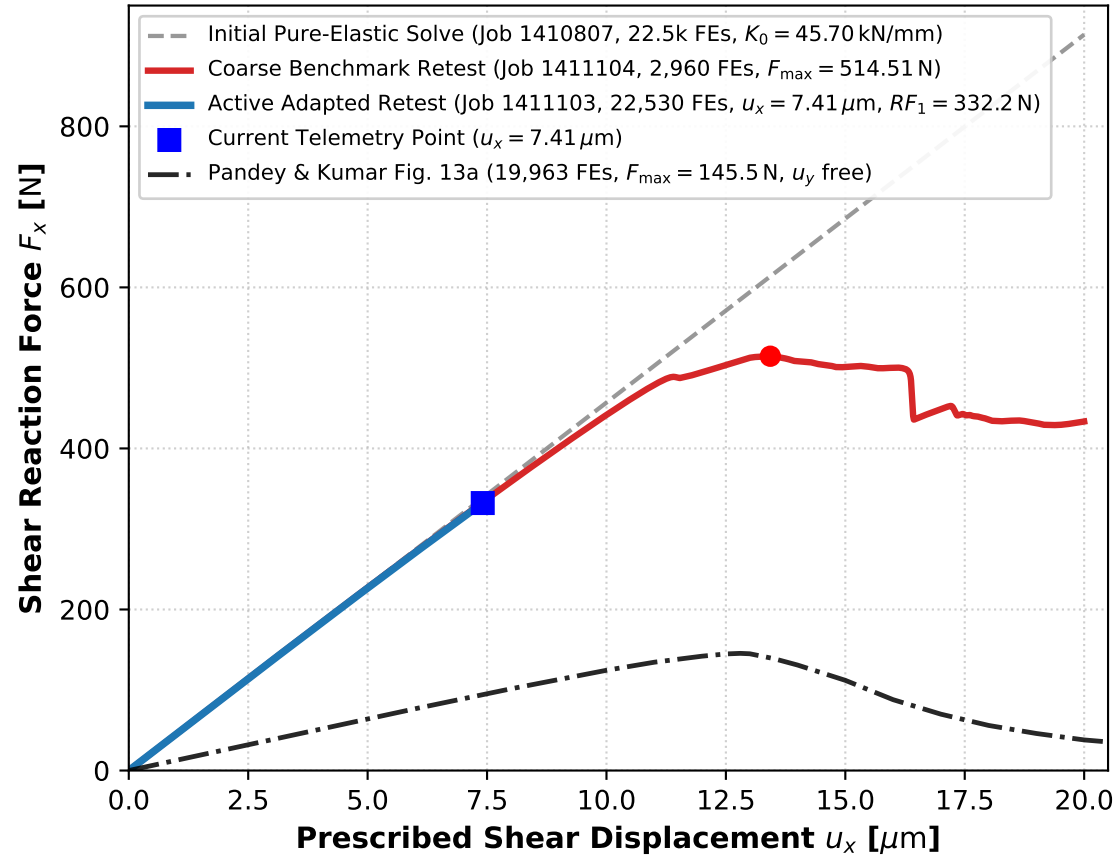
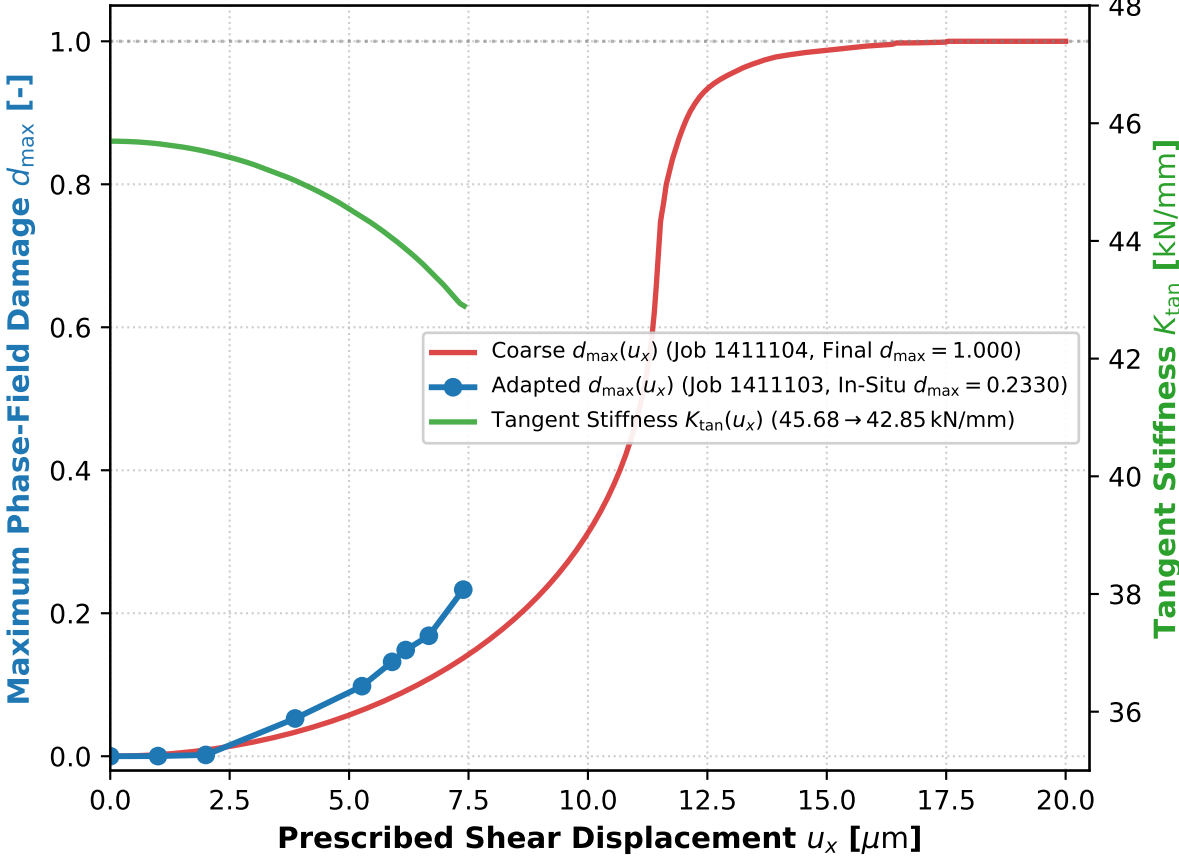


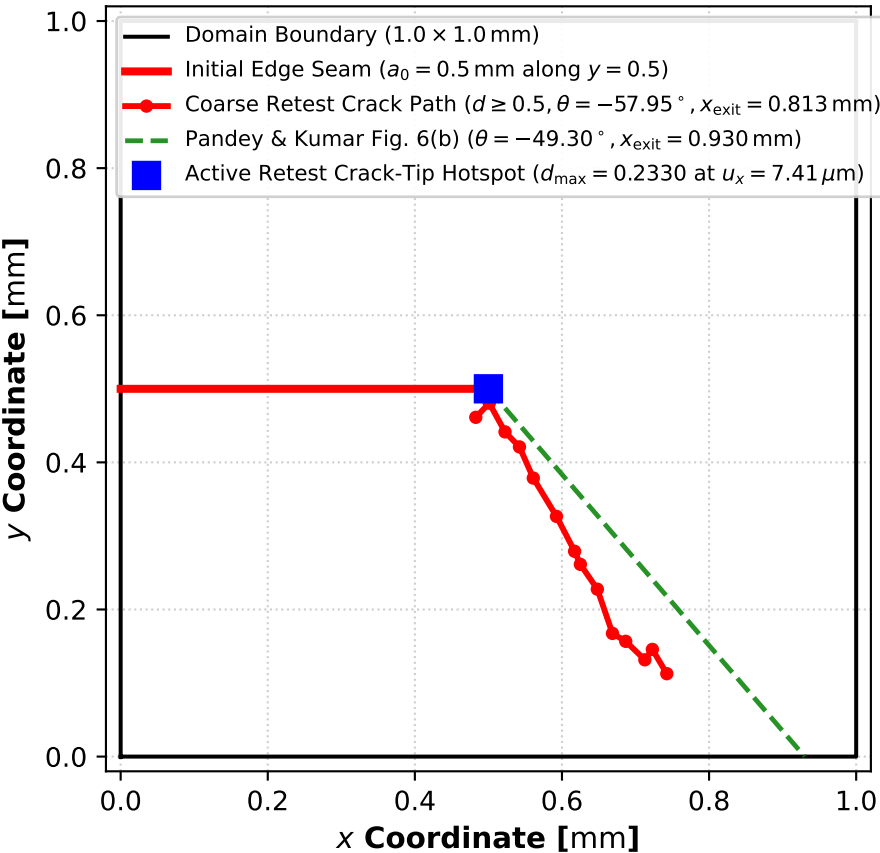
(a) Global Force-Displacement Response & Solver Lineage



(b) Phase-Field Damage Growth & Stiffness Softening



(c) Computed Mode-II Crack Trajectory vs Published Path



(d) Mode-II Multi-Job Provenance & Solver Verification Matrix

PBS Job ID	Job Name	Mesh / FEs	Status / Exit	K_0 [kN/mm]	Peak F [N]	$d_{\max}$	Scientific Role & Lineage
1410790	M2_J1_MIEHE	2,960 FEs	Exit 0 (4,000 incs)	45.80	N/A (Pre-Gate 000)	0.002	Coarse Pre-Analysis Anchor
1410797	M2_J2_ADAPT	22,530 FEs	Exit 0 (Diagnosed)	45.70	913.91	0.006	Initial Defect Offset Isolated & Repaired
1410807	M2_J2_ADAPT	22,530 FEs	Exit 0 (Diagnosed)	45.70	913.91	0.006	Plasticity Validated; RHS Defect Pinpointed
1411104	M2_J1_COARS	2,960 FEs	Exit 0 (Completed)	45.80	514.51	1.000	Companion Coarse Fracture Retest
1411103	M2_J2_RETEST	22,530 FEs	RUNNING (Inc 148)	45.68	332.2	0.2330	Authoritative Fracture Retest